LAB ASSIGNMENT-1

TEAM MEMBERS:

1. Anuj Saha (202351010)
2. Divyanshu Ghosh (202351036)
3. G Nikhil (202351037)

The solution to the given problem is addressed using the encoder-decoder system. Their functions are mentioned below:

1. **Encoder**
    - Accepts an input image.
    - Applies spatial sampling and intensity quantization.
    - Stores the processed image along with a compact header in a custom binary file.
2. **Decoder**
    - Read the custom binary file.
    - Extracts resolution information from the header.
    - Reconstructs and displays the decoded image.

How to Run the Program

1. Place `encoder.m`, `decoder.m` and an input image in the same folder.
2. Run the `encoder.m`.
3. Then
    - Select the image.
    - Choose spatial and intensity resolutions.
    - The encoded binary file is generated.
4. Run the `decoder.m`
    - The decoded image is reconstructed and displayed.

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1. Encoder.m

```
clc;
clear;
close all;
%% Taking the Image as input
[file, path] = uigetfile({'*.jpg;*.png;*.bmp'}, 'Select an Image');
img = imread(fullfile(path, file));
% Grayscale conversion
if ndims(img) == 3
   R = img(:,:,1);
   G = img(:,:,2);
   B = img(:,:,3);
   img = uint8(0.2989*double(R) + 0.5870*double(G) + 0.1140*double(B));
end
figure;
imshow(img);
title('Input Grayscale Image');
%% Central square cropping
[M, N] = size(img);
side = min(M, N);
rs = floor((M - side)/2) + 1;
cs = floor((N - side)/2) + 1;
img_crop = img(rs:rs+side-1, cs:cs+side-1);
%% Spatial Resolution
disp('Select Spatial Resolution');
disp('0 : 100 x 100');
disp('1 : 200 x 200');
disp('2 : 400 x 400');
disp('3 : 800 x 800');
s_idx = input('Enter choice (0-3): ');
resolutions = [100 200 400 800];
new_size = resolutions(s_idx+1);
step = side / new_size;
img_sampled = zeros(new_size, new_size);
for i = 1:new_size
   for j = 1:new_size
       x = round((i-0.5)*step + 0.5);
       y = round((j-0.5)*step + 0.5);
       img_sampled(i,j) = img_crop(x,y);
   end
end
img_sampled = uint8(img_sampled);
%% Intensity Resolution
disp('Select Intensity Resolution');
disp('0 : 1 bit (2 levels)');
disp('1 : 2 bits (4 levels)');
disp('2 : 4 bits (16 levels)');
disp('3 : 8 bits (256 levels)');
```

```
i_idx = input('Enter choice (0-3): ');
bit_depths = [1 2 4 8];
bits = bit_depths(i_idx+1);
levels = 2^bits;
img_quant = zeros(new_size, new_size);
for i = 1:new_size
   for j = 1:new_size
       p = img_sampled(i,j);
       q = floor(double(p) * levels / 256);
       img_quant(i,j) = q * (256 / levels);
   end
end
img_quant = uint8(img_quant);
%% Header
% [ S1 S0 I1 I0 ]
header = s_idx * 4 + i_idx;   % fits in 4 bits (0-15)
%% Making the binary file
fid = fopen('encoded_image.bin', 'wb');
fwrite(fid, header, 'uint8');         % Header
fwrite(fid, new_size, 'uint16');      % Image size
fwrite(fid, img_quant(:), 'uint8');   % Pixel data (linear stream)
fclose(fid);
disp('Encoding complete. Custom binary file created.');
```

2. Decoder.m

```
clc;
clear;
close all;
%% Reading the binary file
fid = fopen('encoded_image.bin', 'rb');
if fid == -1
   error('encoded_image.bin not found. Run encoder.m first.');
end
resolutions = [100 200 400 800];
bit_depths = [1 2 4 8];
% REading the header
header = fread(fid, 1, 'uint8');
decoded_s = floor(header / 4);
decoded_i = mod(header, 4);
decoded_res = resolutions(decoded_s+1);
decoded_bits = bit_depths(decoded_i+1);
img_size = fread(fid, 1, 'uint16');
img_data = fread(fid, img_size * img_size, 'uint8');
decoded_img = reshape(img_data, img_size, img_size);
decoded_img = uint8(decoded_img);
fclose(fid);
%% Displaying the final image
disp(['Decoded Spatial Resolution : ', num2str(decoded_res), ' x ',
num2str(decoded_res)]);
disp(['Decoded Intensity Depth    : ', num2str(decoded_bits), ' bits']);
figure;
imshow(decoded_img);
title('Decoded Image');
```

Results

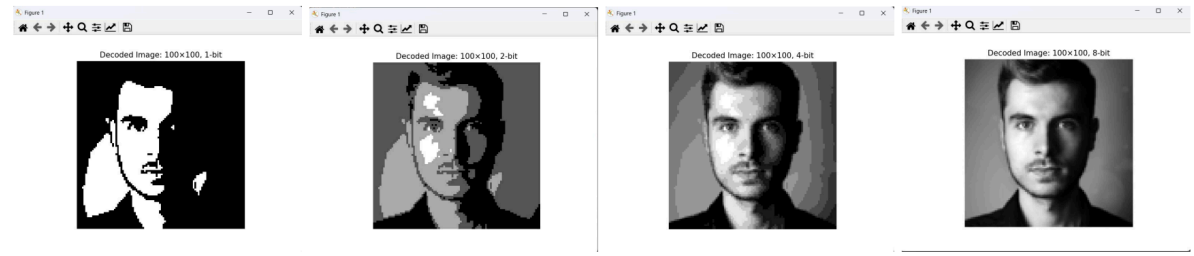

Figure: 100x100 Resolution for 1bit, 2bit, 4bit and 8bit respectively

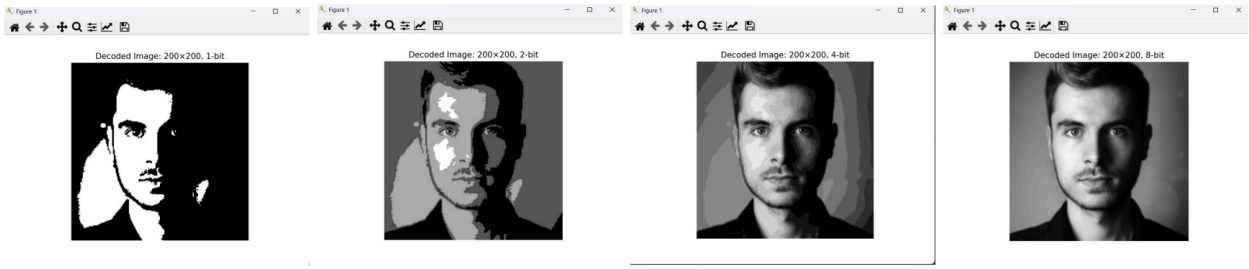

Figure: 200x200 Resolution for 1bit, 2bit, 4bit and 8bit respectively

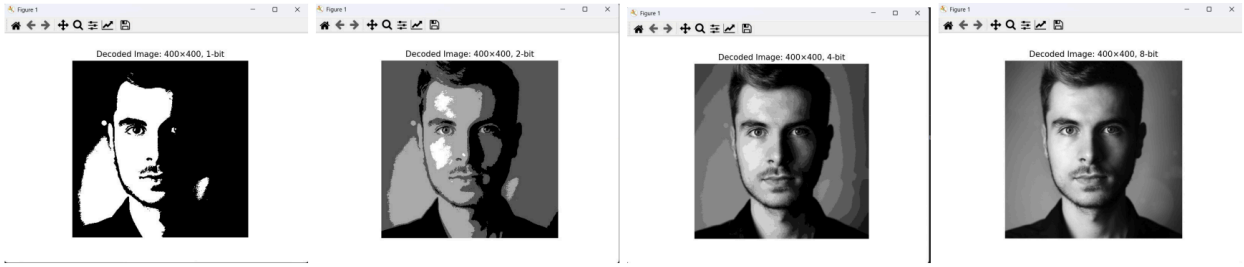

Figure: 400x400 Resolution for 1bit, 2bit, 4bit and 8bit respectively

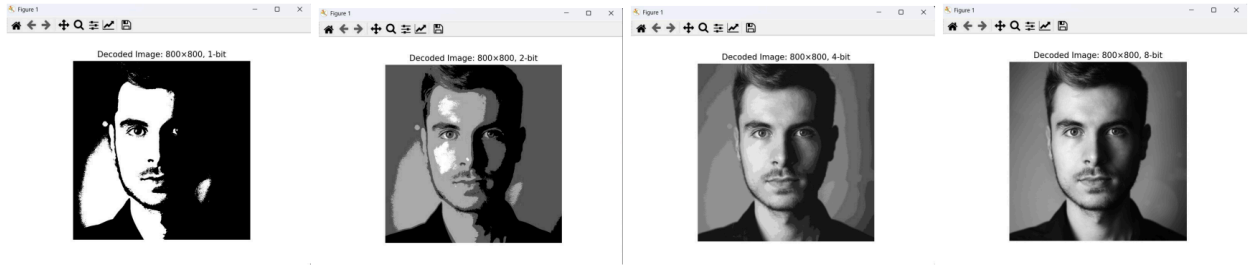

Figure: 800x800 Resolution for 1bit, 2bit, 4bit and 8bit respectively